

AI-Augmented QA Agent

A Jira-to-Playwright QA assistant that converts ticket context into test strategy, automation coverage insight, pending Playwright scripts, and execution evidence.

Primary Audience

QA Engineers, SDETs, QA Leads, and delivery teams.

Core Value

Faster QA planning, clearer risk visibility, and improved automation readiness.

1. Project Title

AI-Augmented QA Agent

2. Problem Statement

QA engineers spend significant time reading Jira tickets, understanding acceptance criteria, identifying testing risks, preparing manual test cases, checking whether automation already exists, and collecting test execution evidence. This work is often repetitive, inconsistent across tickets, and difficult to scale when multiple releases, environments, and QA assignees are involved.

The challenge is to reduce manual preparation time while keeping QA judgment, human approval, and governance in the process.

3. What The Solution Does

AI-Augmented QA Agent is a local QA engineering platform that connects Jira ticket intelligence with Playwright automation. It helps a QA engineer move from ticket review to actionable testing outputs in one guided workflow.

- Loads scoped Jira tickets and filters them by QA assignee.
- Analyzes ticket details, comments, labels, components, and acceptance criteria.
- Generates a QA summary, risk classification, assumptions, and dependencies.
- Creates manual test strategy and test case suggestions.
- Checks Playwright coverage against existing tagged tests.
- Identifies missing or partial automation coverage.
- Generates pending Playwright scripts for human review.
- Runs tagged Playwright suites and supports HTML, JSON, screenshot, video, and trace evidence.

4. How It Works

- 1 Load Jira scope:** The local API server connects to Jira Cloud and retrieves tickets from the configured QA boards and statuses.
- 2 Select ticket:** The React UI displays the scoped ticket list and allows filtering by QA assignee, such as Sana.khan or Naveed Chughtai.

- 3 **Normalize context:** Ticket title, description, acceptance criteria, comments, labels, components, status, and assignee data are prepared for analysis.
- 4 **Analyze QA risk:** The analysis service generates a concise QA summary, risk level, assumptions, dependencies, and testing focus areas.
- 5 **Generate test strategy:** The system suggests manual test cases covering positive, negative, regression, and edge scenarios.
- 6 **Check automation coverage:** Acceptance criteria are compared with existing Playwright tags and specs to identify covered, partial, and missing areas.
- 7 **Suggest automation:** For uncovered scenarios, a pending Playwright spec can be generated into the pending test area.
- 8 **Require approval:** Generated scripts stay pending until a human reviews and approves them before adoption into the main framework.
- 9 **Execute and review:** QA can run tagged Playwright suites locally or through CI and review reports, screenshots, videos, traces, and JSON results.

5. Tools And Technologies Used



6. Practical Example / Use Case

Example: Testing a Jira ticket for checkout or authentication changes

A QA engineer selects a Jira ticket assigned for testing. The agent reads the ticket details and produces a QA testing summary, identifies risks such as login dependency, payment flow impact, environment dependency, or missing acceptance criteria, and then generates manual test cases.

Next, it checks whether existing Playwright tests already cover related areas such as **@login**, **@auth**, **@cart**, or **@checkout**. If coverage is missing, it drafts a pending Playwright spec for review. After approval, QA can run the relevant tagged suite and attach the generated report evidence to the delivery process.

7. Key Benefits And Time-Saving Impact

- **Faster ticket understanding:** Reduces time spent manually reading and summarizing Jira tickets.
- **Consistent QA planning:** Produces repeatable test strategy, risks, assumptions, and dependencies for each ticket.
- **Improved automation awareness:** Highlights what is already covered and what still needs automation.
- **Reduced duplicate effort:** Helps avoid writing new tests for scenarios already covered by existing Playwright tags.
- **Better review governance:** Keeps generated scripts pending until approved by a human.

- **Stronger release evidence:** Supports Playwright reports, screenshots, videos, traces, and JSON outputs for review.
- **Portfolio and team value:** Demonstrates a practical SDET workflow that can be extended for real team adoption.

8. Future Improvements

- Add OpenAI structured-output powered ticket analysis.
- Add PR-based approval flow for generated Playwright specs.
- Generate AI-style execution summaries from Playwright JSON reports.
- Add flaky test detection, retry analytics, and test impact analysis from changed files.
- Add Slack or Microsoft Teams notification summaries.
- Add role-based approval controls for generated automation.
- Expand Jira custom field mapping and release dashboard metrics.

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